# MILITARY SPECIFICATION

## MICROCIRCUITS, LINEAR, VOLTAGE REGULATOR, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, voltage regulator. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

| Device type | Circuit |
|---|---|
| 01 | Precision voltage regulator |

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

| Outline letter | Case outline (see MIL-M-38510, appendix C) |
|---|---|
| A | F-1 (14-lead, 1/4" x 1/4"), flat package |
| B | F-3 (14-lead, 3/16" x 1/4"), flat package |
| C | D-1 (14-lead, 1/4" x 3/4"), dual-in-line package |
| D | F-2 (14-lead, 1/4" x 3/8"), flat package |
| H | F-4 (10-lead, 1/4" x 1/4"), flat package |
| I | A-2 (10-lead can) |
| 2 | C-2 (20-terminal, .350" x .350"), square chip carrier package |

1.3 Absolute maximum ratings.

| | |
|---|---|
| Pulse voltage from V+ to V- (50 ms) | 50 V |
| Continuous voltage from V+ to V- | 40 V |
| Input-output voltage differential | 40 V |
| Differential input voltage | ±5 V |
| Voltage between noninverting input and V- | +8 V |
| Current from $V_Z$ | 25 mA |
| Current from $V_{REF}$ | 15 mA |
| Lead temperature (soldering, 10 seconds) | 300°C |
| Junction temperature ($T_J$) 1/ | +175°C |
| Storage temperature range | -65°C to 150°C |

1/ For short term test (up to 100 hours, maximum) $T_J$ = 275°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.


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1.4 Recommended operating conditions.

| | |
|---|---|
| Input voltage range | 9.5 to 40 V dc |
| Output voltage range | 2 to 37 V dc |
| Input-output voltage differential | 2.5 to 38 V dc |
| Ambient operating temperature range ($T_A$) | -55°C to +125°C |

1.5 Power and thermal characteristics.

| Package | Case outline | Maximum allowable power dissipation | Maximum $\theta_{JC}$ | Maximum $\theta_{JC}$ |
|---|---|---|---|---|
| 14-lead FP | A, B, D | 350 mW @ $T_A$ = 125°C | 60°C/W | 140°C/W |
| Dual-in-line | C | 400 mW @ $T_A$ = 125°C | 35°C/W | 120°C/W |
| 10-lead FP | H | 350 mW @ $T_A$ = 125°C | 60°C/W | 150°C/W |
| 10-lead can | I | 350 mW @ $T_A$ = 125°C | 40°C/W | 140°C/W |
| 20-terminal CCP | 2 | 72 mW @ $T_A$ = 125°C | 55°C/W | 121°C/W |

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the these documents shall be those listed in the issue of the Department of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.



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3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.4 Typical schemtic circuit. The typical schematic circuit shall be as specified on figure 2.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range.

$$V_{IN} = V^+ = V_{CC} = 12 \text{ V dc}$$

TABLE I. Electrical performance characteristics.

| Characteristics | Symbol | Conditions 1/ 2/ | | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Line regulation | $V_{RLINE}$ | $V_{IN}$ = 12 to 15 V<br>$V_{OUT}$ = 5 V,<br>$I_L$ = 1 mA | $T_A$ = +25°C | -0.1 | 0.1 | % $V_{OUT}$ |
| | | | $T_A$ = -55°C | -0.3 | 0.3 | % $V_{OUT}$ |
| | | | $T_A$ = +125°C | -0.2 | 0.2 | % $V_{OUT}$ |
| | $V_{RLINE}$ | $V_{IN}$ = 9.5 to 40 V<br>$V_{OUT}$ = 5 V,<br>$I_L$ = 1 mA | $T_A$ = +25°C | -0.3 | 0.3 | % $V_{OUT}$ |
| | $V_{RLINE}$ | $V_{OUT}$ = 2 V,<br>$I_L$ = 1 mA<br>$V_{IN}$ = 12 to 40 V | $T_A$ = +25°C | -0.2 | 0.2 | % $V_{OUT}$ |
| Load regulation | $V_{RLOAD}$ | $I_L$ = 1 to 50 mA<br>$V_{OUT}$ = 5 V,<br>$V_{IN}$ = 12 V | $T_A$ = +25°C | -0.15 | 0.15 | % $V_{OUT}$ |
| | | | $T_A$ = -55°C | -0.6 | 0.6 | % $V_{OUT}$ |
| | | | $T_A$ = +125°C | -0.4 | 0.4 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | $V_{IN}$ = 40 V,<br>$I_L$ = 1 to 10 mA<br>$V_{OUT}$ = 37 V | $T_A$ = +25°C | -0.5 | 0.5 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | $V_{IN}$ = 10 V,<br>$I_L$ = 6 to 12 mA<br>$V_{OUT}$ = 7.5 V | $T_A$ = +25°C | -0.2 | 0.2 | % $V_{OUT}$ |
| Reference voltage | $V_{REF}$ | $I_{REF}$ = 1 mA<br>$V_{IN}$ = 12 V | $T_A$ = +25°C | 6.95 | 7.35 | V dc |
| | | | -55°C ≤ $T_A$ = 125°C | 6.90 | 7.40 | V dc |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Characteristics | Symbol | Conditions 1/ 2/ | | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|
| Output short circuit current | $I_{OS}$ | $R_{SC}$ = 10Ω, $R_L$ = 0<br>$V_{OUT}$ = 5 V,<br>$V_{IN}$ = 12 V | $T_A$ = +25°C | 45 | 85 | mA |
| Standby current drain | $I_{SCD}$ | $I_L = I_{REF}$ = 0,<br>$V_{IN}$ = 30 V<br>$V_{OUT} = V_{REF}$ | $T_A$ = +25°C | 0.5 | 3.0 | mA |
| | | | $T_A$ = -55°C | 0.5 | 3.5 | mA |
| | | | $T_A$ = +125°C | 0.5 | 2.4 | mA |
| Average temperature coefficient of output voltage | $TC_{VOUT}$ | $V_{OUT}$ = 5 V,<br>$V_{IN}$ = 12 V<br>$I_L$ = 1 mA | +25°C ≤ $T_A$ ≤ +125°C | -.010 | 0.010 | %/°C |
| | | | -55°C ≤ $T_A$ ≤ -55°C | -.015 | 0.015 | %/°C |
| Zener voltage | $V_Z$ | Available in packages A, B, C, D, and 2 only | $T_A$ = +25°C<br>$I_Z$ = 1 mA | 5.58 | 6.82 | V dc |
| Ripple rejection | $\frac{\Delta V_{OUT}}{\Delta V_{IN}}$ | f = 50 Hz to 10 kHz, see figure 3<br>$V_{OUT}$ = 5 V | $C_{REF}$ = 0 $T_A$ = +25°C | 64 | | dB |
| | | | $C_{REF}$ = 5 µF $T_A$ = +25°C | 76 | | dB |
| Output noise | $N_O$ | BW = 100 Hz to 10 kHz, see figure 3<br>$V_{OUT}$ = 5 V | $C_{REF}$ = 0 $T_A$ = +25°C | | 120 | µV rms |
| | | | $C_{REF}$ = 5 µF $T_A$ = +25°C | | 7.0 | µV rms |
| Line transient response | $\frac{\Delta V_{OUT}}{\Delta V_{IN}}$ | $V_{IN}$ = 12 V, $V_{OUT}$ = 5 V, $I_L$ = 1 mA, $T_A$ = +25°C<br>$R_{SC}$ = 0Ω, $\Delta V_{IN}$ = 3 V for 25 µs, see figure 4 | | 0 | 10 | mV/V |
| Load transient response | $\frac{\Delta V_{OUT}}{\Delta I_L}$ | $V_{IN}$ = 12 V, $V_{OUT}$ = 5 V, $I_L$ = 40 mA, $T_A$ = +25°C<br>$R_{SC}$ = 0Ω, $\Delta I_L$ = 10 mA for 25 µs, see figure 4 | | -1.5 | 0 | mV/mA |

1/ All $V_{OUT}$ values are nominal.

2/ All regulation requirements are based on a constant junction temperature.



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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests in table II after part number marking, to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

| MIL-STD-883 test requirements | Subgroups (see table III) 1/ | |
|---|---|---|
| | Class S devices | Class B devices |
| Interim electrical parameters (method 5004) | 1 | 1 |
| Final electrical test parameters (method 5004) | 1, 2, 3 | 1, 2, 3 |
| Group A test requirements (method 5005) | 1, 2, 3, 4, 8 | 1, 2, 3, 8 |
| Group C end-point and group B class S electrical parameters (method 5005) | 1, 2, 3 and table IV delta limits | Table IV delta limits and limits |
| Additional electrical subgroups for group C periodic inspections | N/A | 4 |
| Group D end-point electrical parameters (method 5005) | 1, 2, 3 | 1 |

1/ PDA applies to subgroup 1 (see 4.2.d).



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TABLE III. Group A inspection.

| Subgroup | Symbol | Test no. | Conditions 1/ 2/ | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|
| 1<br>$T_A$ = +25°C | $V_{RLINE}$ | 1 | $V_{OUT}$ = 5 V, $I_L$ = 1 mA<br>$V_{IN}$ = 12 to 15 V | -0.10 | 0.10 | % $V_{OUT}$ |
| | $V_{RLINE}$ | 2 | $V_{OUT}$ = 5 V, $I_L$ = 1 mA<br>$V_{IN}$ = 9.5 to 40 V | -0.3 | 0.3 | % $V_{OUT}$ |
| | $V_{RLINE}$ | 3 | $V_{OUT}$ = 2 V, $I_L$ = 1 mA<br>$V_{IN}$ = 12 to 15 V | -0.2 | 0.2 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | 4 | $V_{OUT}$ = 5 V, $V_{IN}$ = 12 V<br>$I_L$ = 1 to 50 mA | -0.15 | 0.15 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | 5 | $V_{OUT}$ = 37 V, $V_{IN}$ = 40 V<br>$I_L$ = 1 to 10 mA | -0.5 | 0.5 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | 6 | $V_{OUT}$ = 7.5 V, $V_{IN}$ = 10 V<br>$I_L$ = 6 to 12 mA | -0.2 | 0.2 | % $V_{OUT}$ |
| | $V_{REF}$ | 7 | $V_{IN}$ = 12 V<br>$I_{REF}$ = 1 mA | 6.95 | 7.35 | V dc |
| | $I_{OS}$ | 8 | $R_{SC}$ = 10Ω, $R_L$ = 0<br>$V_{OUT}$ = 5 V, $V_{IN}$ = 12 V 3/ | 45 | 85 | mA |
| | $I_{SCD}$ | 9 | $I_L$ = $I_{REF}$ = 0, $V_{IN}$ = 30 V,<br>$V_{OUT}$ = $V_{REF}$ | 0.5 | 3.0 | mA |
| | $V_Z$ | 10 | $I_Z$ = 1 mA<br>Available in packages A, B, C, D, and 2 only | 5.58 | 6.82 | V dc |
| 2<br>$T_A$ = +125°C | $V_{RLINE}$ | 11 | $V_{OUT}$ = 5 V, $I_L$ = 1 mA<br>$V_{IN}$ = 12 to 15 V | -0.2 | 0.2 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | 12 | $V_{OUT}$ = 5 V, $V_{IN}$ = 12 V<br>$I_L$ = 1 to 50 mA | -0.4 | 0.4 | % $V_{OUT}$ |
| | $V_{REF}$ | 13 | $I_{REF}$ = 1 mA, $V_{IN}$ = 12 V | 6.90 | 7.40 | V dc |
| | $I_{SCD}$ | 14 | $I_L$ = $I_{REF}$ = 0, $V_{IN}$ = 30 V,<br>$V_{OUT}$ = $V_{REF}$ | 0.5 | 2.4 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection - Continued.

| Subgroup | Symbol | Test no. | Conditions 1/ 2/ | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|
| 3<br>$T_A$ = -55°C | $V_{RLINE}$ | 15 | $V_{OUT}$ = 5 V, $I_L$ = 1 mA<br>$V_{IN}$ = 12 to 15 V | -0.3 | 0.3 | % $V_{OUT}$ |
| | $V_{RLOAD}$ | 16 | $V_{OUT}$ = 5 V, $V_{IN}$ = 12 V<br>$I_L$ = 1 to 50 mA | -0.6 | 0.6 | % $V_{OUT}$ |
| | $V_{REF}$ | 17 | $I_{REF}$ = 1 mA, $V_{IN}$ = 12 V | 6.90 | 7.40 | V dc |
| | $I_{SCD}$ | 18 | $I_L$ = $I_{REF}$ = 0, $V_{IN}$ = 30 V,<br>$V_{OUT}$ = $V_{REF}$ | 0.5 | 3.5 | mA |
| 4<br>$T_A$ = +25°C | Ripple rej | 19 | f = 10 kHz, $C_{REF}$ = 0, see figure 3 | 64 | | dB |
| | Ripple rej | 20 | f = 10 kHz, $C_{REF}$ = 5 µF, see figure 3 | 76 | | dB |
| | Output noise | 21 | BW = 100 Hz to 10 kHz, $C_{REF}$ = 0,<br>see figure 3 | | 120 | µV rms |
| | Output noise | 22 | BW = 100 Hz to 10 kHz, $C_{REF}$ = 5 µF,<br>see figure 3 | | 7.0 | µV rms |
| | Line transient response | 23 | $V_{IN}$ = 12 V, $V_{OUT}$ = 5 V, $I_L$ = 1 mA,<br>$C_{REF}$ = 5 µF,<br>$R_{SC}$ = 0Ω, $\Delta V_{IN}$ = 3 V, see figure 4 | 0 | 10 | mV/V |
| | Load transient response | 24 | $V_{IN}$ = 12 V, $V_{OUT}$ = 5 V, $I_L$ = 40 mA,<br>$C_{REF}$ = 5 µF,<br>$R_{SC}$ = 0Ω, $\Delta I_L$ = 10 mA, see figure 4 | -1.5 | 0 | mV/mA |
| 8<br>$T_A$ = +125°C | TC$V_{OUT}$ | 25 | $V_{OUT}$ = 5 V, $V_{IN}$ = 12 V<br>$I_L$ = 1 mA | -0.010 | 0.010 | %/°C |
| 8<br>$T_A$ = -55°C | TC$V_{OUT}$ | 26 | $V_{OUT}$ = 5 V, $V_{IN}$ = 12 V<br>$I_L$ = 1 mA | -0.015 | 0.015 | %/°C |

1/ All $V_{OUT}$ values are nominal.

2/ Unless otherwise specified, use figure 5 .

3/ To eliminate heating, test must be made in less than 10 ms, duty cycle of less than 5 percent. $V_{OUT}$ is the nominal output voltage prior to application of short circuit.



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TABLE IV. Group C, end point electrical parameters ($T_A$ = 25°C).

| Table III test no. | Characteristic | Symbol | Delta limits 1/ | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|
| 1 | Line regulation | $V_{RLINE}$ | ±15% or 1 mV 2/ | -0.10 | 0.10 | % $V_{OUT}$ |
| 4 | Load regulation | $V_{RLOAD}$ | ±20% or 1 mV 2/ | -0.15 | 0.15 | % $V_{OUT}$ |
| 7 | Reference voltage | $V_{REF}$ | ±15 mV | 6.95 | 7.35 | V dc |
| 9 | Standby current drain | $I_{SCD}$ | ±10% | 0.5 | 3.0 | mA dc |

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

2/ Whichever is greater.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D) using the circuit shown on figure 6; test duration shall be 240 hours minimum. Static tests (test condition A) using the circuit shown on figure 7; test duration shall be 48 hours minimum.

(2) Class B devices: Test condition A using the circuit shown on figure 7; or test condition D using the circuit shown on figure 6.

b. Reverse bias burn-in (method 1015 of MIL-STD-883) shall be omitted.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).



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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 7. If the alternate burn-in conditions are used, the circuit shown on figure 6 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady-state life test class B (method 1005 of MIL-STD-883): Test condition C using the circuit shown on figure 6 or test condition F using the circuit shown on figure 7.

c. A special subgroup shall be added to group C inspection and shall consist of the tests, conditions, and limits specified for group A subgroup 4 as specified in table III herein. The LTPD for the special subgroup shall be 5 for all classes.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and end-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

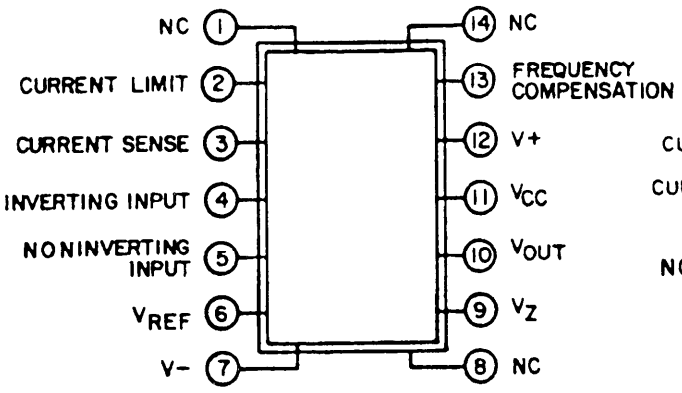
4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

4.5.2 Life test and burn-in cool down procedure. When device are measured at 25°C following application of steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to the removal of the bias.

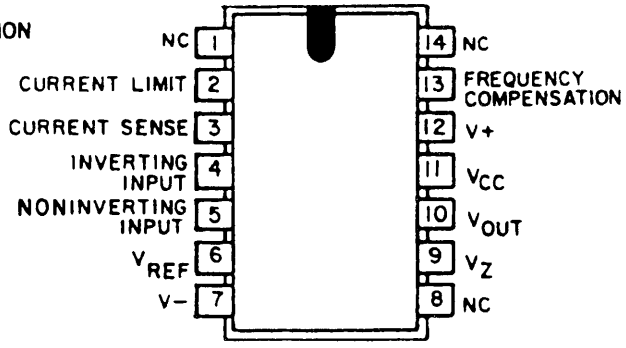


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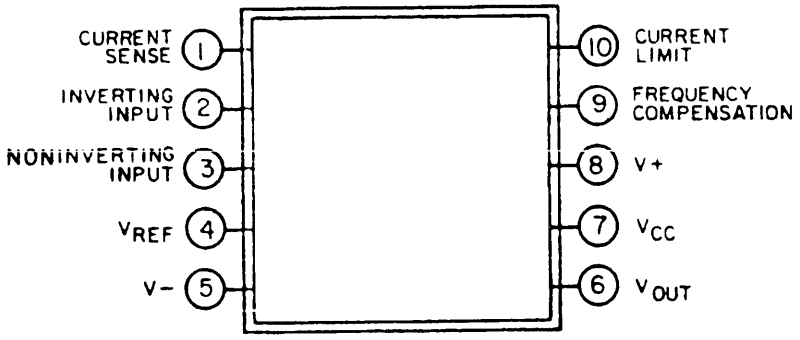
Cases A, B and D




Case C




Case H




Case I

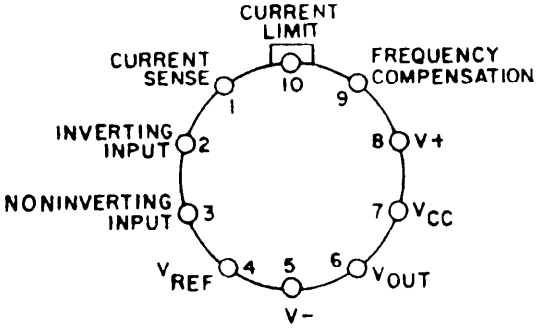
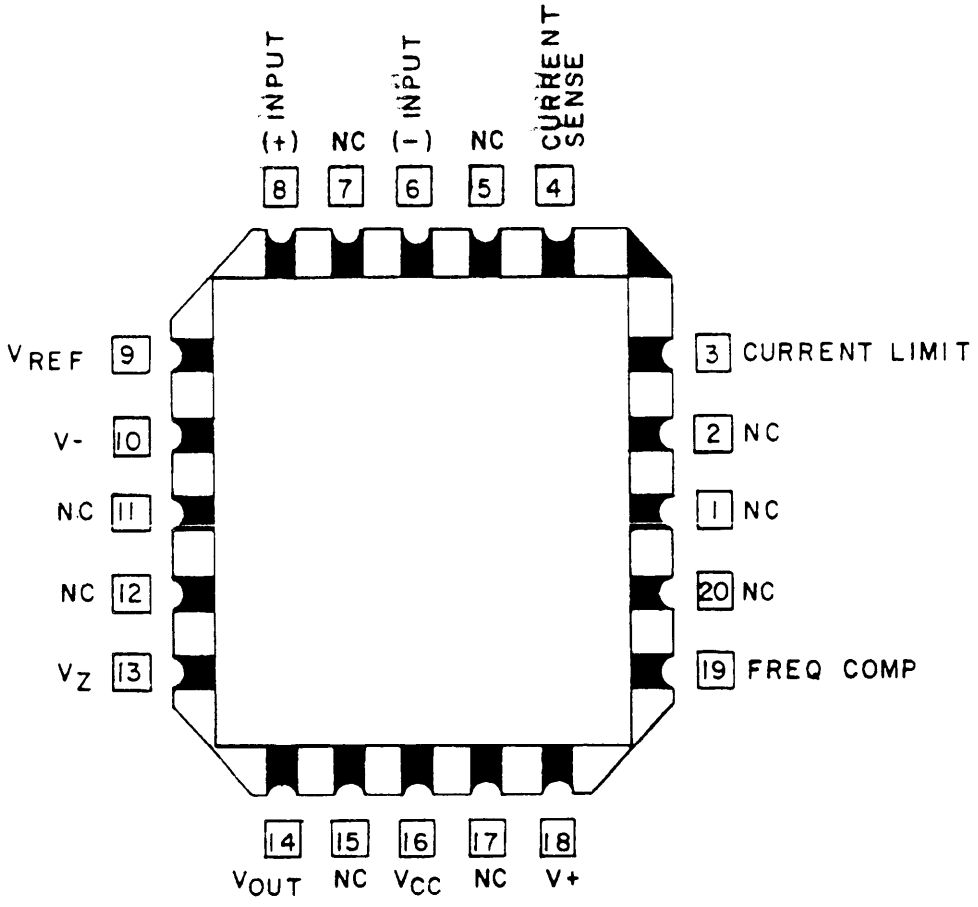



FIGURE 1. Terminal connections.



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Case 2



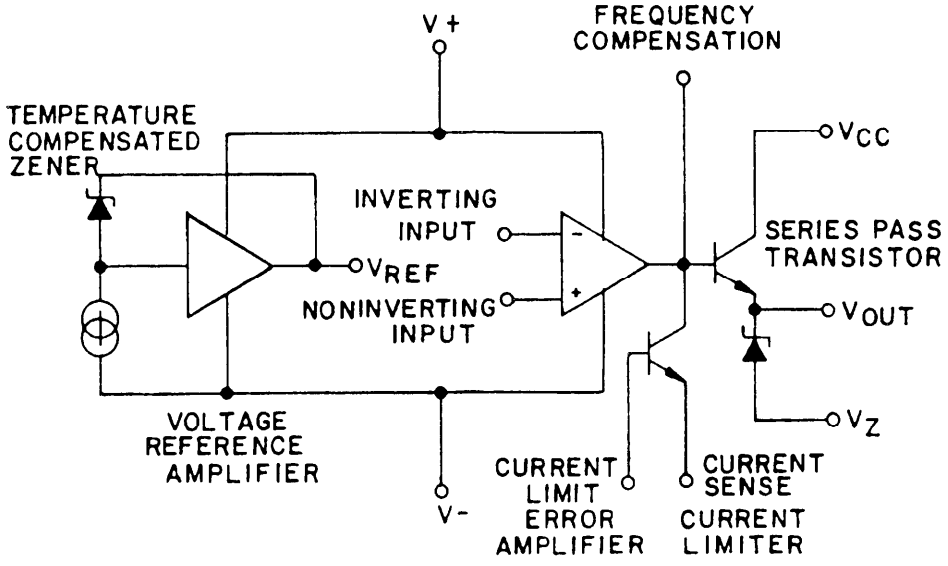

NOTE: V- connected to case of metal packages.

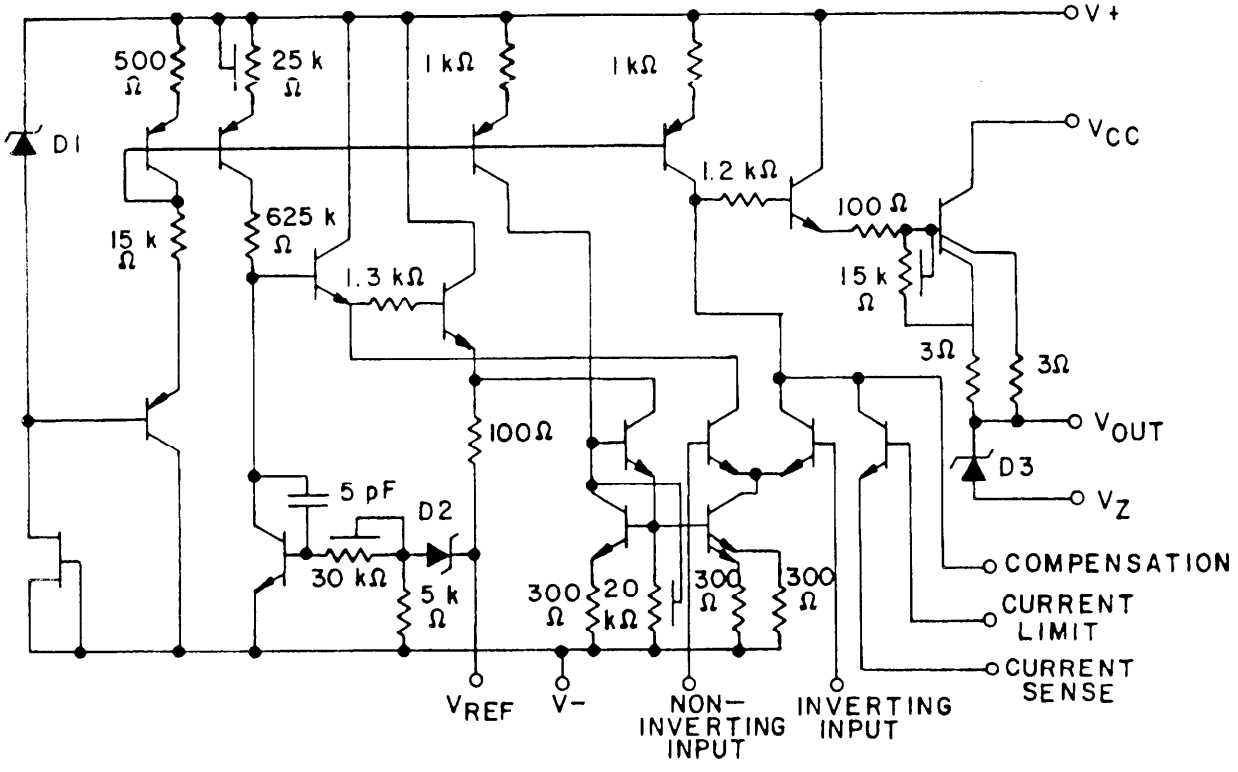
FIGURE 1. Terminal connections - Continued.



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Block diagram



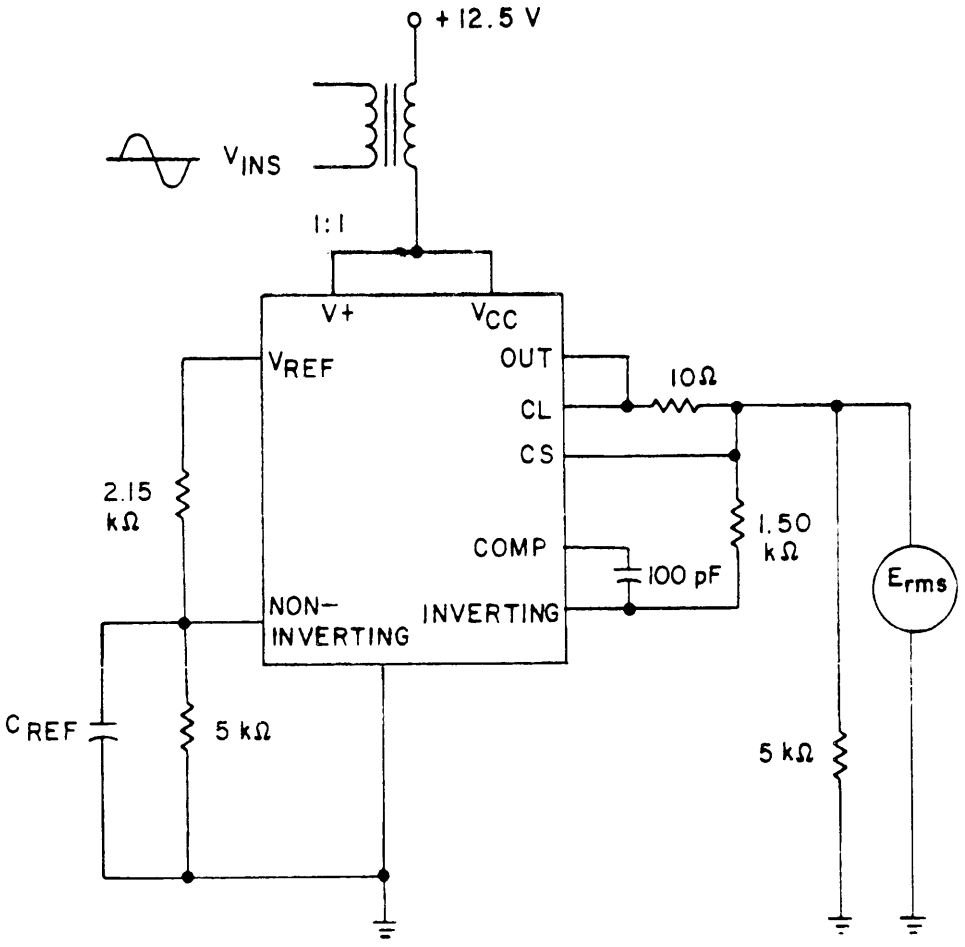




NOTE: All resistance and capacitance values are nominal.

FIGURE 2. Typical schematic circuit.



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NOTES:

1. For ripple rejection
   $V_{INS}$ = 2 V rms, 10 kHz

$$\text{Ripple rejection (db)} = 20 \log \frac{V_{INS}}{E_{rms}}$$

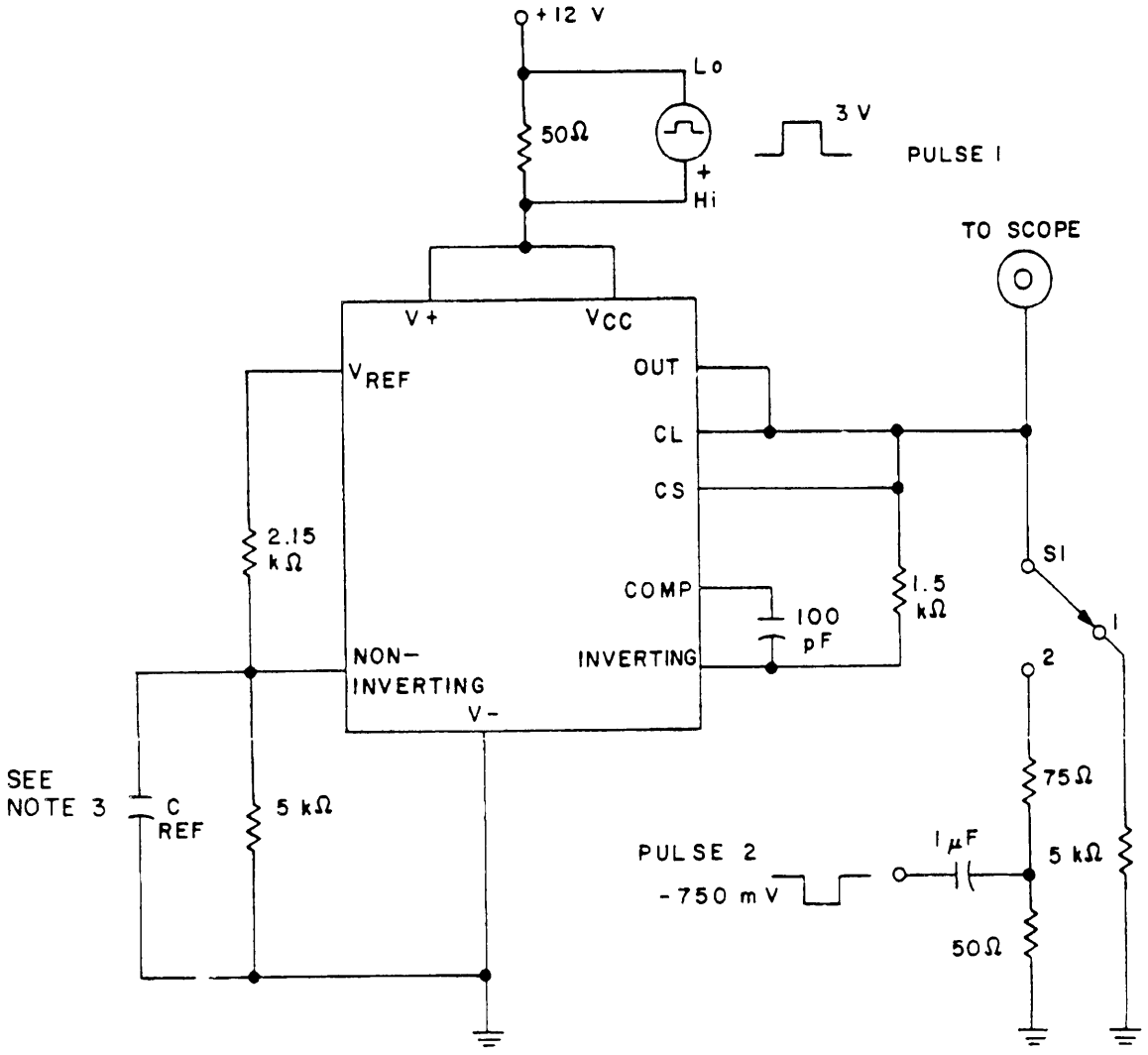
2. For noise
   $V_{INS}$ = 0 V rms
   $E_{rms}$ is over a band width of 100 Hz to 10 kHz = noise V rms.

3. $C_{REF}$ is as specified in table III.

FIGURE 3. Ripple rejection and noise test circuit.



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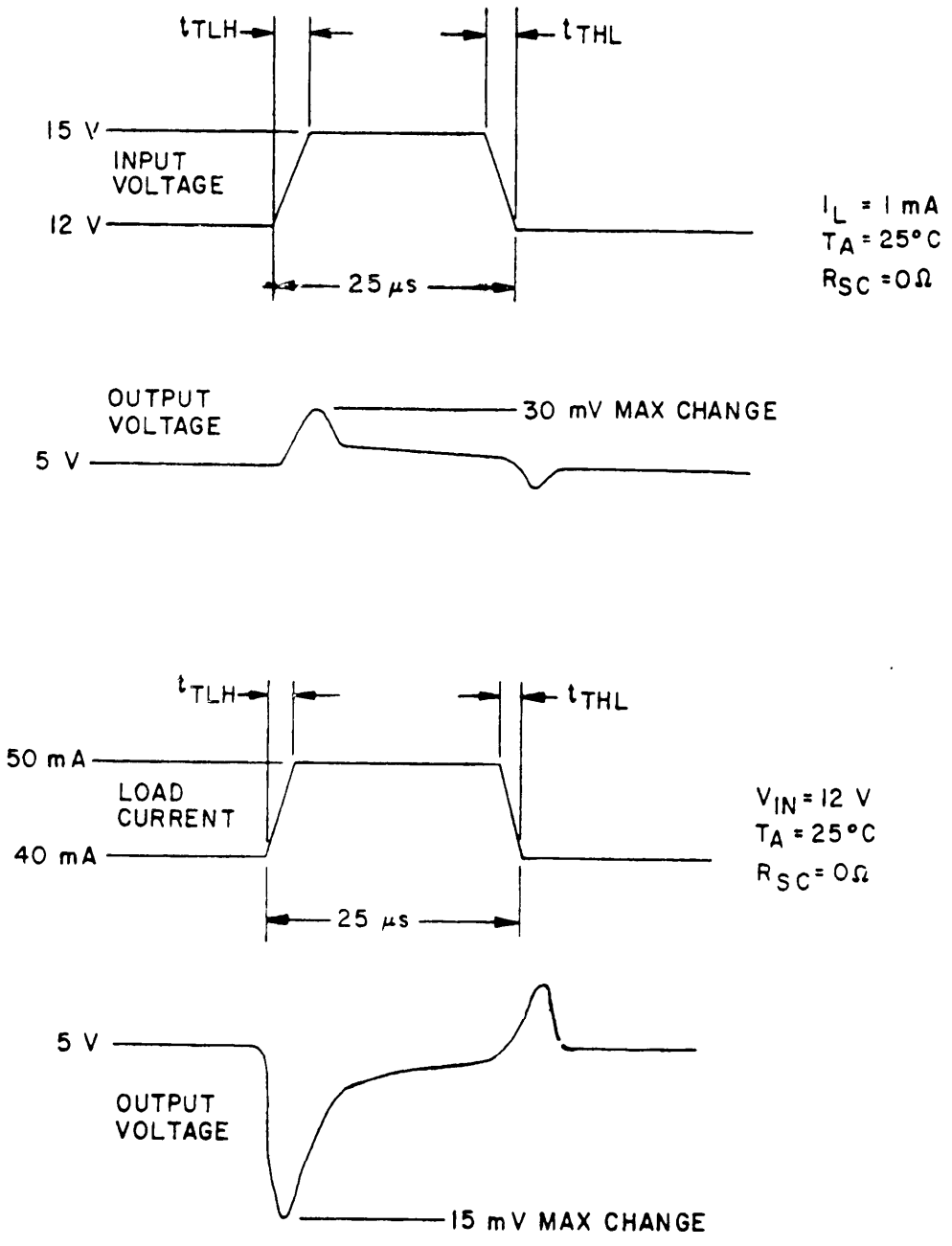



NOTES:
1. For line transient response, $S_1$ in position 1
   pulse 1 equals 3 V high, 25 μs wide, 3 percent duty cycle.
   $t_{TLH} = t_{THL} = 1$ μs.
2. For load transient response, S1 in position 2
   pulse 1 = 0
   pulse 2 equals -750 mV, 25 μs wide, 3 percent duty cycle,
   $t_{TLH} = t_{THL} = 1$ μs.
3. $C_{REF}$ is as specified in table III.

FIGURE 4. Transient response test circuit.



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Load transient response

FIGURE 4. Transient response test circuit - Continued.



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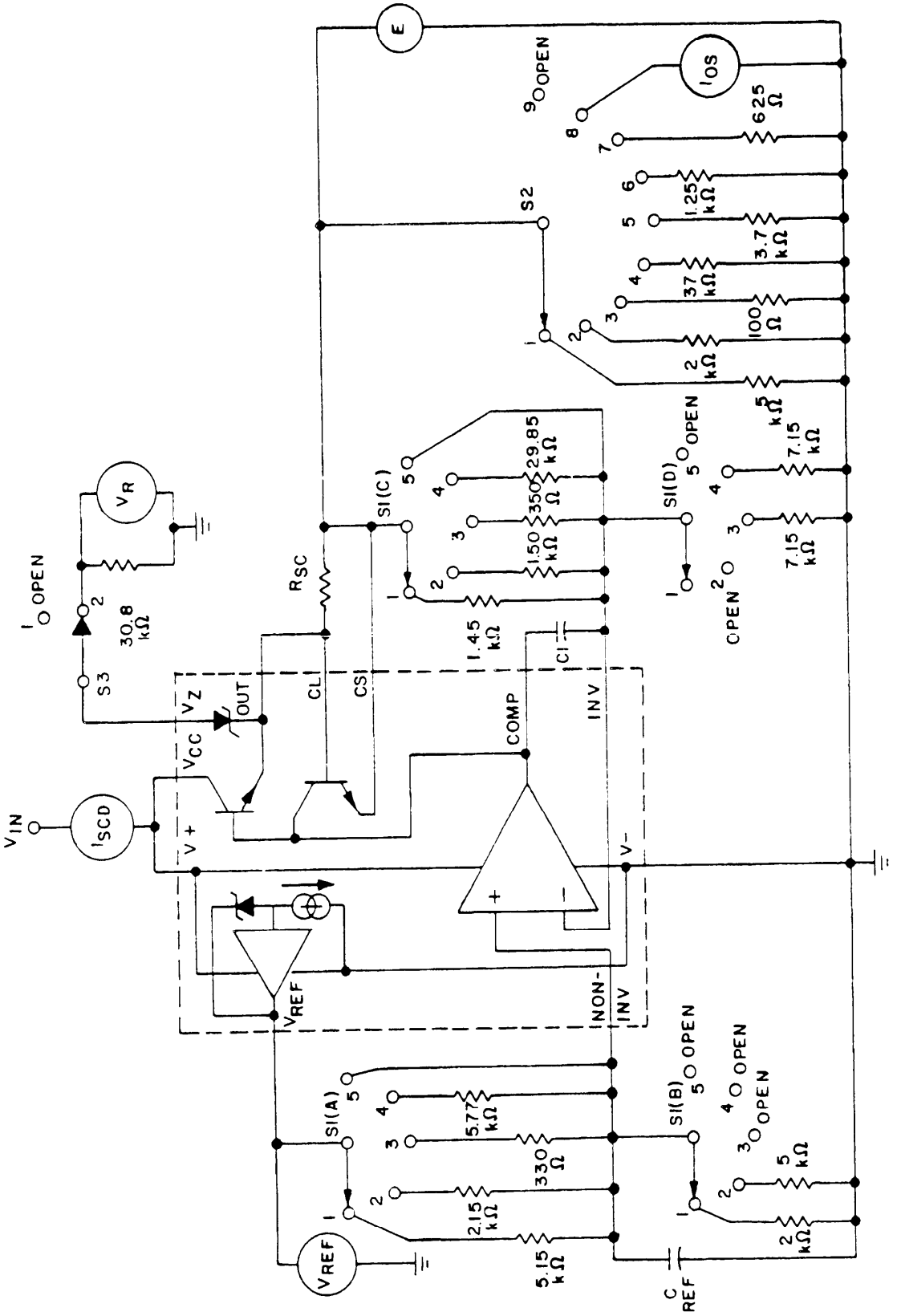



FIGURE 5. Test circuit for static and dynamic tests.



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| Parameter | Test | $V_{IN}$(volts) | Switch positions S1 | S2 | S3 | Measure Value | Units | Measured parameters equation | Units |
|---|---|---|---|---|---|---|---|---|---|
| $V_{RLINE}$ | 1, 11, 15 | 12 / 15 | 2 / 2 | 1 / 1 | 1 / 1 | E1 / E2 | Volts / Volts | $V_{RLINE} = \frac{E1 - E2}{E1} \times 100$ | $\Delta V_{OUT}$ |
| $V_{RLINE}$ | 2 | 9.5 / 40 | 2 / 2 | 1 / 1 | 1 / 1 | E3 / E4 | Volts / Volts | $V_{RLINE} = \frac{E3 - E4}{E3} \times 100$ | $\Delta V_{OUT}$ |
| $V_{RLINE}$ | 3 | 12 / 15 | 1 / 1 | 2 / 2 | 1 / 1 | E5 / E6 | Volts / Volts | $V_{RLINE} = \frac{E5 - E6}{E5} \times 100$ | $\Delta V_{OUT}$ |
| $V_{RLOAD}$ | 4, 12 16 | 12 / 12 | 2 / 2 | 1 / 3 | 1 / 1 | E7 / E8 | Volts / Volts | $V_{RLOAD} = \frac{E7 - E8}{E7} \times 100$ | $\Delta V_{OUT}$ |
| $V_{RLOAD}$ | 5 | 40 / 40 | 4 / 4 | 4 / 5 | 1 / 1 | E9 / E10 | Volts / Volts | $V_{RLOAD} = \frac{E9 - E10}{E9} \times 100$ | $\Delta V_{OUT}$ |
| $V_{RLOAD}$ | 6 | 10 / 10 | 3 / 3 | 6 / 7 | 1 / 1 | E11 / E12 | Volts / Volts | $V_{RLOAD} = \frac{E11 - E12}{E11} \times 100$ | $\Delta V_{OUT}$ |
| $V_{REF}$ | 7, 13, 17 | 12 | 2 | 1 | 1 | $V_{REF}$ | Volts | | |
| $I_{OS}$ | 8 | 12 | 2 | 8 | 1 | $I_{OS}$ | mA | | |
| $I_{SCD}$ | 9, 14, 18 | 30 | 5 | 9 | 1 | $I_{SCD}$ | mA | | |
| $V_Z$ | 10 | 40 | 4 | 4 | 2 | $V_R$ | Volts | $V_Z = E9 - V_R$ | Volts |
| $TC_{VOUT}$ | 26, 27 | 12 | 2 | 1 | 1 | E1 | Volts | See notes 4 and 5 | %/°C |

FIGURE 5. Test circuit for static and dynamic tests - Continued.



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NOTES:

1. Unless otherwise specified:
   $V_{IN} = V^+ = V_{CC} = 12$ V, V- = 0 V, $I_L = 1$ mA, $R_{SC} = 0\Omega$, C1 = 100 pF and $C_{REF} = 0$.

2. Resistor values required to set output voltage.

   a. For $V_{OUT}$ = 2 to 7 V dc, R1, R2, R3, and R4 are determined as follows:

   $$V_{OUT} = V_{REF} \times \frac{R2}{R1 + R2} \qquad \frac{V_{REF}}{R1 + R2} = 1 \text{ mA}$$

   $$\frac{R1 \quad R2}{R1 + R2} = R3 \leq 10 \text{ k}\Omega \qquad R4 = \text{Open circuit}$$

   b. For $V_{OUT}$ = 7 to 37 V dc, R1, R2, R3, and R4 are determined as follows:

   $$V_{OUT} = V_{REF} \times \frac{R3 + R4}{R4} \qquad \frac{V_{OUT}}{R3 + R4} = 1 \text{ mA}$$

   $$\frac{R3 \quad R4}{R3 + R4} = R1 \leq 10 \text{ k} \qquad R2 = \text{Open circuit}$$

   c. For the purpose of this specification, the following table shall be used to determine the resistor value required to obtain a given nominal output voltage.

| | R1 | R2 | R3 | R4 |
|---|---|---|---|---|
| $V_{OUT}$ | S1(A) | S1(B) | S1(C) | S1(D) |
| 2 V | 5.15 k | 2 k | 1.45 k | ∞ |
| 5 V | 2.15 k | 5 k | 1.50 k | ∞ |
| 7.5 V | 0.33 k | ∞ | 0.35 k | 7.15 k |
| 37 V | 5.77 k | ∞ | 29.85 k | 7.15 k |

   If 1 percent resistors are used, nominal $V_{OUT}$ will be achieved within ±5 percent.

3. $R_L$ chosen to obtain required $I_L$ for given nominal output voltage.

4. $$T_C V_{OUT} \text{ 25°C to 125°C} = 100\left(\frac{V_{OUT} @ 25°C - V_{OUT} @ 125°C}{V_{OUT} @ 25°C}\right) \div 100°C = \%/°C$$

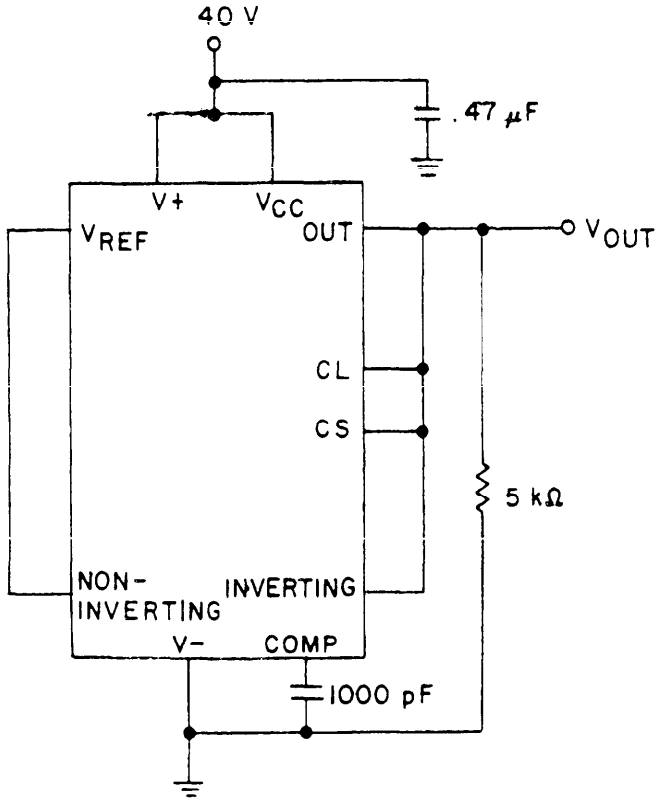
5. $$T_C V_{OUT} \text{ -55°C to 25°C} = 100\left(\frac{V_{OUT} @ 55°C - V_{OUT} @ 25°C}{V_{OUT} @ 25°C}\right) \div 100°C = \%/°C$$

6. Precautions shall be taken to prevent damage to the DUT during insertion into socket and change of switch positions (e.g., disable power supplies, current limit, $V_{CC}$, V+, V-, etc.).

FIGURE 5. Test circuit for static and dynamic tests - Continued.



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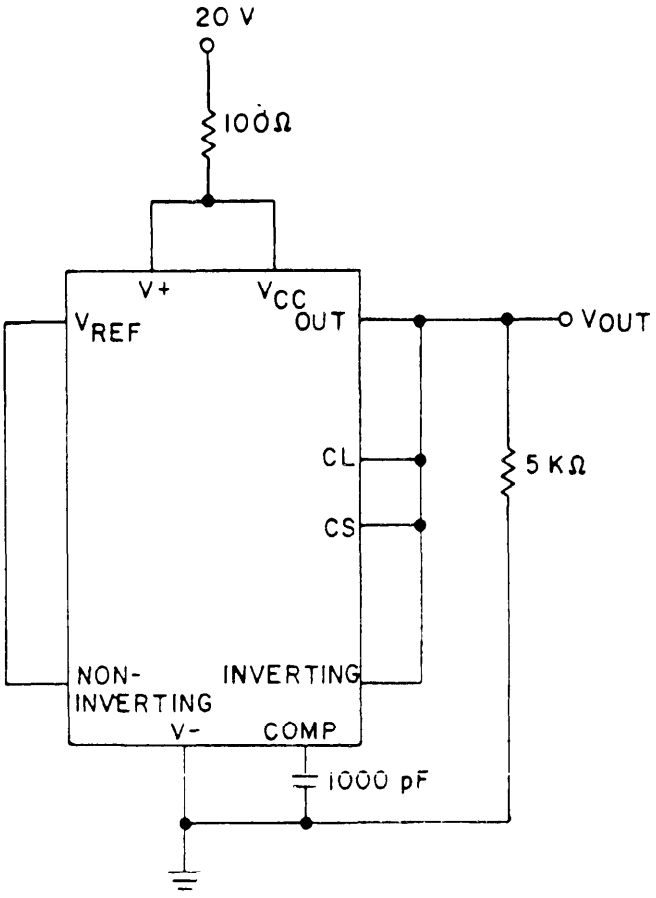

Pin number designations

| Case | CS | INV | N. INV | $V_{REF}$ | V- | Out | $V_{CC}$ | V+ | Comp | CL |
|---|---|---|---|---|---|---|---|---|---|---|
| H and I | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A, B, C, and D | 3 | 4 | 5 | 6 | 7 | 10 | 11 | 12 | 13 | 2 |
| 2 | 4 | 6 | 8 | 9 | 10 | 14 | 16 | 18 | 19 | 3 |

FIGURE 6. Burn-in, steady state life test, and reverse bias test circuit.



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Pin number designations

| Case | CS | INV | N. INV | $V_{REF}$ | V- | Out | $V_{CC}$ | V+ | Comp | CL |
|---|---|---|---|---|---|---|---|---|---|---|
| H and I | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A, B, C, and D | 3 | 4 | 5 | 6 | 7 | 10 | 11 | 12 | 13 | 2 |
| 2 | 4 | 6 | 8 | 9 | 10 | 14 | 16 | 18 | 19 | 3 |

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device during test at high temperature. The device is not considered functional under accelerated test conditions.

FIGURE 7. Accelerated burn-in and life test circuit.



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for JAN marking.

6.3 Abbreviations, symbols, and definitions. Unless otherwise defined, the abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

Line regulation. The percentage change in output voltage for a specified change in input voltage ($V_{RLINE}$).

Load regulation. The percentage change in output voltage for a specified change in load current ($V_{RLOAD}$).

Ripple rejection. The ratio of the peak to peak input ripple voltage to the peak to peak output ripple voltage ($\Delta V_{OUT}/\Delta V_{IN}$).

Average temperature coefficient of output voltage. The percentage change in output voltage for a specified change in ambient temperature ($TC_{VOUT}$).

Short circuit current limit. The output current of the regulator with the output shorted to the negative supply ($I_{OS}$).

Reference voltage. The output of the reference amplifier measured with respect to the negative supply ($V_{REF}$).

Output noise voltage. The rms output noise voltage with constant load and no input ripple ($N_O$).

Standby current drain. The supply current drawn by the regulator with no output load and no reference voltage load ($I_{SCD}$).



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Input voltage range. The range of supply voltage over which the regulator will operate.

Input-output voltage differential. The range of voltage difference between the supply voltage and the regulated output voltage over which the regulator will operate.

Sense voltage. The voltage between current sense and current limit terminals necessary to cause current limiting.

Transient response. The closed-loop step function response of the regulator under small-signal conditions.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

| Military device type | Generic industry type |
|---|---|
| 01 | 723 |

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC
Air Force - 80

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-38510/102B | Military Specification Microcircuits, Linear, Voltage Regulator, Monolithic Silicon |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
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